

Spatiotemporal-Aware Deep Reinforcement Learning for Multi-UAV Cooperative Coverage in Emergency Deterministic Communications

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Abstract—In post-disaster reconstruction and rescue work, unmanned aerial vehicles (UAVs) are often used as aerial base stations to provide communication services for ground users (GUs), owing to their flexibility and controllability. The premise for the efficient completion of cooperative rescue tasks is the deterministic transmission of data, that is, to ensure that the end-to-end delay of data (Time Sensitive Flow, TSF) is bounded and the jitter is stable. However, a single UAV is insufficient to fully cover the entire target area. To obtain an efficient multi-UAV cooperative control coverage strategy that directs multiple UAVs to maximize the number of coverage tasks in a cooperative manner, the centralized controller must jointly consider the spatial positions of both the mission and the UAVs, as well as the effective transmission time windows for TSFs. Towards this end, we propose a spatiotemporal aware deep reinforcement learning (STA-DRL) method. The proposed method 1) The transformer is used to extract spatiotemporal features from environmental states to improve the generalization representation ability of the model; 2) Prioritized Experience Replay Buffer helps accelerate model convergence; 3) A coverage strategy is formulated under the guidance of two Transformer models. Simulation results show that the proposed method is superior to existing methods in terms of task completion coverage quantity.

Index Terms—UAV, time sensitive flow, deterministic communication coverage, deep reinforcement learning.

I. INTRODUCTION

RECENTLY, natural disasters have frequently occurred around the world. Disaster-affected areas that rely mainly on base stations for communication services are often unable to operate due to damage to the power system or base stations,

which poses a significant challenge to rapid post-disaster response and reconstruction efforts [1]. Drones are often used to build post-disaster emergency communication networks due to their flexibility and controllability [2], [3]. Existing emergency communication networks generally use a single drone as an aerial base station to provide access and relay communications to the ground. However, the network system capacity and communication coverage area of a single drone are relatively limited and it is not able to provide long-distance remote communications to GUs. In order to address these limitations, inspired by swarm intelligence, researchers began to form multiple unmanned aerial vehicles (UAVs) into a UAV network to provide communication services for a large area, so as to improve the system capacity and communication coverage area of the network [4].

To address the time-critical nature of post-disaster emergency rescue, technologies [5] such as telemedicine surgery and autonomous rescue robots have been developed, all of which depend on reliable and timely communication networks to operate effectively. When deployed promptly, these technologies can significantly reduce casualties among disaster-affected populations. However, the premise for realizing these technologies is to ensure that the sensor data generated by them can be delivered to the destination node completely and on time, which is also the deterministic communication requirement of time sensitive flows (TSFs) [6], [7]. Fig. 1 shows that a UAV network composed of multiple UAVs provides relay transmission services to ground user nodes in the earthquake area with external networks (air-ground integrated emergency communication system), in which the cluster head (CH) serves as a relay node between the cluster member (CM) and the external network. For example, a mobile hospital in the earthquake area needs to perform telemedicine surgery from 9:40 to 10:20. The rescue command center must deploy drones to provide communication coverage for the mobile hospital before the mission begins and establish a reliable communication link between the mobile hospital and the telemedicine center. During the operation, the UAV provides stable link quality for data transmission by hovering until the end of the operation, then may proceed to other coverage tasks.

A further analysis of the situation reveals that deterministic communication coverage (DCC) mainly involves two key considerations. First, if the deployed UAV fails to provide timely communication coverage for the mobile hospital, the telemedicine center will be unable to perform telemedicine

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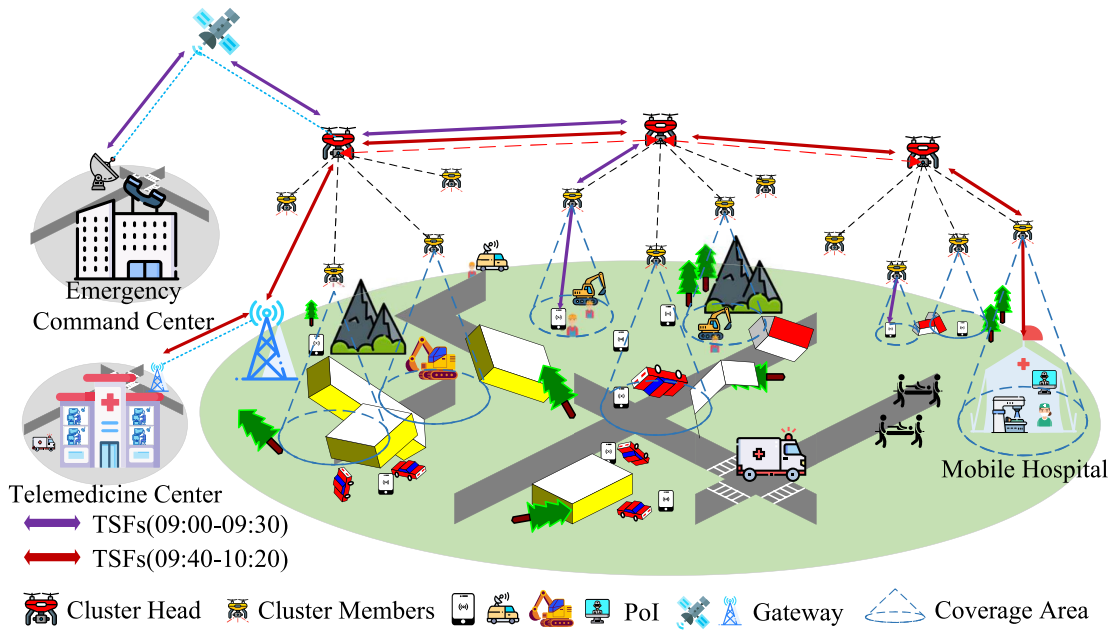


Fig. 1. A UAV network providing deterministic communication coverage for GUs in a target region.

surgery on the victims. Second, during the operation, any interruption in communication between the mobile hospital and the telemedicine center, caused by drone movement, will reduce the success rate of the procedure. Compared to existing IP network communication mechanisms, deterministic communication methods with temporal certainty are better suited for application scenarios that require the provision of access services to users in disaster areas at precise time points.

In the emergency deterministic communication scenario, the DCC problem of multiple drones can be modeled as a markov decision process with spatiotemporal characteristics (ST-MDP), aimed at maximizing the number of successful task completions. However, there are two significant challenges in solving this problem. First, given the limited number and coverage range of UAVs, along with the time-sensitive nature of coverage tasks, taking into account the temporal sequence and intervals between tasks facilitates a reasonable allocation of UAVs to each task. Second, the spatial dependency between task locations within the target area influences the spatial path planning of the drones. In summary, it is a considerable challenge to use a finite number of drones to provide DCC for multiple time-sensitive tasks in a large target area.

To address this challenge, we propose a deep reinforcement learning algorithm with spatiotemporal awareness for efficient multi-UAV communication coverage. This algorithm leverages the unique feature extraction capabilities of deep learning [8], [9], [10], [11] and the outstanding performance of reinforcement learning in UAV path planning (UPP) tasks. Once the emergency command center receives a task requiring communication support (location and time), it is necessary to plan the paths of all drones based on the tasks and the number of available drones. To optimize the number of tasks covered, we introduce a deep reinforcement learning (DRL) approach where Transformer is employed as the Q-value function approximator in the deep

Q-network (DQN). The spatial dependencies between UAVs and tasks (spatial attention features), as well as the temporal dependencies between tasks (temporal attention features), are extracted using multi-head attentions. These features are then cascaded into spatiotemporal attention features and processed through a softmax layer to probabilistically select the most suitable action for each UAV. Key experiences are stored using prioritized experience replay. During training, the target Q-network is updated with a soft update mechanism. This method not only accelerates the convergence of the model, but also improves the robustness of the model. The main contributions of this work are summarized as follows.

- 1) Since the environmental state contains key spatial and temporal information, the multi-UAV cooperative coverage problem is modeled as a markov decision process with spatiotemporal characteristics, which enables the centralized controller to efficiently develop a cooperative coverage strategy.
- 2) Considering both the spatial dependency between UAVs and tasks in the environment, as well as the temporal dependency between tasks, the attention mechanism is employed to learn the spatiotemporal features of the environment, thereby improving the model's robustness.
- 3) Consequently, a spatiotemporal-aware deep reinforcement learning algorithm based on the transformer is proposed. This algorithm is designed to adaptively assign coverage tasks to individual UAVs. Numerical results demonstrate that, compared to existing approaches, the proposed scheme effectively meets the deterministic transmission requirements for TSFs and improves the number of successful task completions.

The remainder of this paper is structured as follows. Section II reviews related work on existing UPP methods. Section III presents the system model and formulates the

problem. In Section IV, we detail the principles of the proposed STA-DRL approach. Section V presents the experimental results, and Section VI concludes the paper.

II. RELATED WORKS

Existing UAV emergency communication coverage works mainly focus on relaying data for GUs, providing “best effort” services [12], [13], [14]. Compared with the former, TSFs have QoS requirements such as periodic forwarding, bounded end-to-end transmission delay, and stable jitter. As rescue technology gradually develops towards intelligence, the proportion of such services in post-disaster emergency communications is gradually increasing [15], [16], [17]. However, so far, there is a lack of research on disaster emergency communication networks that support the transmission of TSFs. Inspired by the application of Time Sensitive Networks (TSNs) in the Industrial Internet of Things (IIoTs) [18], [19], and considering the differences between industrial and post-disaster environments, we propose a UAV deterministic network for disaster emergency response.

Considering emergency situations such as disaster relief and public safety missions, existing research on UPP mainly focuses on providing real-time communication relay and sensor data collection for GUs [20], [21], [22]. These works focus on designing UPP algorithms with the optimization goal of improving communication coverage. Researchers plan the flight paths of multiple UAVs to achieve communication coverage of large-scale target areas. The authors in [20] considers the challenges of establishing a centralized controller for drones and the urgency of emergency communications, proposing a collaborative trajectory design method for UAVs based on the enhanced K-means algorithm and the multi-agent PPO (MAPPO) algorithm. The authors in [21] formulate UAV-assisted vehicle network communication as a long sequence decision problem and propose a dense multi-agent reinforcement learning algorithm with good performance in coverage. In [22], the authors combine the tracking pheromone model with multi-agent deep deterministic policy gradient to maximize the coverage efficiency for users in the disaster area. In the research scenarios of the above work, drones are considered capable of covering the target area at any given time, but are not suitable for solving multi-target communication coverage problems with time constraints.

As the freshness of data transmitted by time-sensitive tasks in emergency communications has garnered widespread attention, existing sensor data collection efforts mainly focus on minimizing the total average information freshness [23], aiming to solve the multi-UAV path coverage problem under time constraints. In UAV-assisted IoT systems, to improve data freshness under energy and mission duration constraints, Huang et al. [24] applies an Asynchronous Advantage Actor-Critic (A3C)-based algorithm to address the UPP problem and derive an optimal trajectory. Considering the time constraint of the message, the minimum descent UAV task completion time is taken as the optimization goal. Liu et al. [25] proposed a trajectory optimization algorithm based on modular coordinate descent. However, these works focus only on trajectory optimization when a single

TABLE I
LIST OF MAJOR NOTATIONS

Notation	Explanation
i, N	The index of a CM, the number of CMs.
j, M	The index of a CH, the number of CHs.
t, T	The index of an epoch, the total number of epochs.
$d_i(t), \theta_i(t)$	Flying distance and direction of UAV i at epoch t .
$\bar{v}$	Average speed of the CM.
h, H	The flying heights of CM and CH.
R	The coverage range.
k, K	The index of a PoI, the number of PoIs.
$r(\cdot), L(\cdot)$	Reward function, loss function.
$Q(\cdot), Q'(\cdot)$	Q function, target Q function.
$a_i(t), r_i(t)$	Action and reward at epoch t .
$s_i(t)$	State at epoch t .

UAV covers multiple tasks and cannot address the problem of cooperative path planning for multiple UAVs.

Since the coverage area of a single UAV is relatively limited, it is difficult to solve the problem of multi-target coverage in a wide area. Some researchers have begun to focus on optimizing the cooperative path planning of multi-UAVs. Gao et al. [26] address the urgency of task execution and the constraints of network cost, and propose a heuristic approximation algorithm to optimize UAV task allocation and minimize task coverage completion time. The authors in [27] defined the UPP as a long-time scale problem and used proximal policy optimization (PPO) to solve the optimal trajectory plan for multiple UAVs. To improve the minimum weighted spectral efficiency among multiple UAVs, Qin et al. [8] formulates the UAV trajectory planning problem as a sequential decision-making task and adopts a centralized Soft Actor-Critic (SAC) algorithm to optimize the communication and sensing performance of the UAVs. Although these works consider the time constraints of tasks to some extent, they do not define specific upper and lower bounds on the time constraints for task coverage. On the contrary, in this paper, the UAV must continuously cover the task within the specified effective time to be considered as effective coverage. This condition implies that not only the time constraints between tasks must be considered, but also the spatial constraints between UAVs and tasks. Therefore, we formulate the multi-task DCC problem as an ST-MDP, and propose an STA-DRL algorithm to maximize the number of successfully completed coverage tasks within the bounded task execution period.

III. SYSTEM MODEL AND PROBLEM STATEMENT

We describe the system model and the multi-UAV collaborative coverage problem in this section. Table I summarizes the main notations used in the remainder of the paper.

A. UAV Deterministic Communication System Model

Deterministic communication refers to the ability of a network to provide bounded end-to-end latency, predictable jitter, and highly reliable data transmission. In UAV-assisted emergency communication networks, deterministic networking (DetNet) [18] technologies can ensure reliable communication for data flows, thereby meeting the stringent requirements for time-liness and stability in post-disaster scenarios.

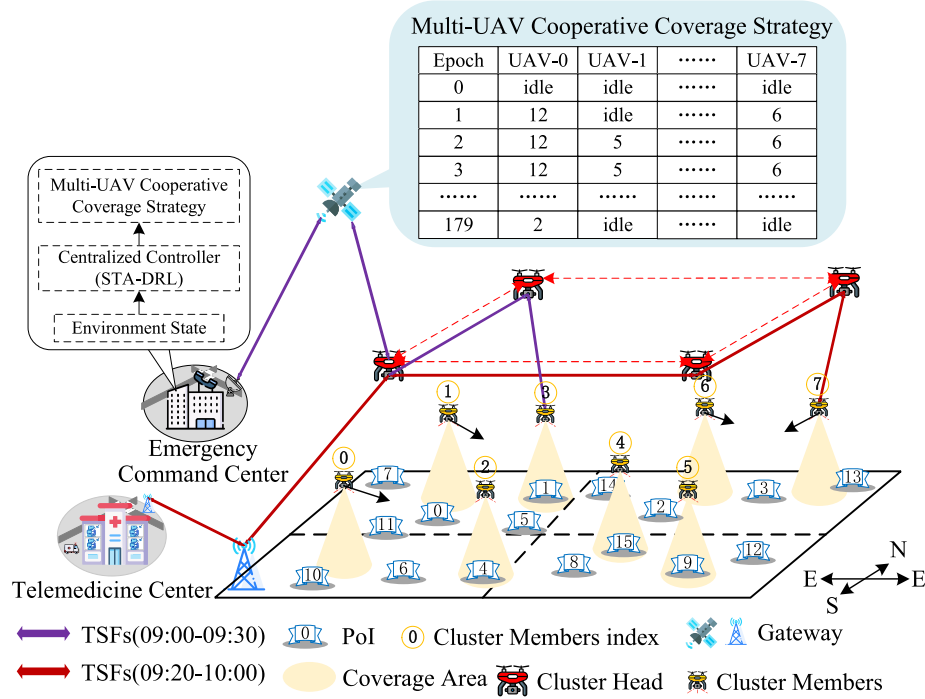


Fig. 2. The UAV deterministic network, where CMs cover PoIs based on multi-UAV cooperative coverage strategy.

We consider a UAV deterministic network comprising a UAV cluster with M large UAVs serving as CHs. The CHs are responsible for relaying data sent by the CMs and distributing the coverage strategies to them. These CHs primarily serve to relay data transmitted by CMs. Additionally, there are N medium-sized UAVs acting as CMs, whose main function is to provide DCC for GUs, referred to as Points of Interest (PoIs), over a target region of size S . The flight altitudes of the CHs and CMs are H and h , respectively, and both are equipped with GPS positioning devices. As shown in Fig. 2, since the communication range of a single CH is limited, it is impossible to establish communication with all CMs in the target region, which is divided into M sub-regions. CMs in each sub-region can communicate with the CH hovering at the center of the sub-region. The CM moves based on the coverage strategy and communicates with the outside world through one or more CHs.

The execution of a multi-UAV cooperative coverage task typically involves three stages over a given task duration T . First, the emergency command center collects all communication coverage requests from GUs that are expected to occur within the time horizon T . Next, a centralized controller processes these requests and generates an optimal cooperative coverage strategy for the UAV network. Finally, each CM executes its assigned coverage tasks in accordance with the coordinated strategy.

Maintaining the stability of the network topology is a prerequisite for establishing a UAV deterministic network [18], [28]. Therefore, a topology control scheme is proposed to improve the stability of UAV networks. The CH, serving as the relay node in the network, is continuously hovered. This ensures that, due to the low probability of interference in the high-altitude wireless channel [29], the quality of the wireless link between

CHs remains relatively stable. In contrast, the CM may interrupt the communication with the CH during the movement. These interruptions, caused by environmental changes, lead to significant fluctuations in the transmission delay of TSF data between CMs and CHs, thus failing to meet the requirements for TSF data transmission. To mitigate the impact of node movement on the wireless link, movement constraints are imposed on CMs during the relaying of TSFs. Specifically, once a TSF is being relayed, the CM must remain stationary until the relay is complete. In summary, this approach ensures that the transmission path for any TSF in the UAV deterministic network remains stable.

B. Movement Model of UAVs

The movement model of UAVs are categorized into two modes: strategy-based flight mode and random walk mode [30]. The strategy-based flight mode guides the CM to move toward a target PoI and perform DCC as planned. Alternatively, the random walk mode is designed to address scenarios where communication may be disrupted by unpredictable signal interference. Upon detecting such interference during communication with a PoI, the CM switches to the random walk mode. While ensuring that the PoI remains within its coverage area, the CM dynamically explores alternative positions to reduce interference and maintain communication quality.

Considering that the initial position of all CMs at the task start time t_0 are $p^{uav}(t_0)$, the moving distance and directions of UAV_i in the epoch t are $d_i(t)$ and $\theta_i(t)$ respectively, and are updated according to the following equation:

$$d_i(t) = \sqrt{\Delta x^2 + \Delta y^2}, \quad (1)$$

$$\theta_i(t) = \arctan\left(\frac{\Delta y}{\Delta x}\right). \quad (2)$$

Therefore, the position of UAV_{*i*} at the *t*-th epoch is $p_i^{uav}(t) = [(x_i(t-1) + \Delta x), (y_i(t-1) + \Delta y)]$.

In this paper, the flight speed of all CMs is fixed at a constant value $\bar{v}$, as illustrated in Fig. 2, the cooperative coverage strategy table defines the coverage target for each CM at every epoch, indicating the specific PoI that each UAV is scheduled to approach and cover. For example, at epoch $t = 0$, UAV-0 is assigned an *idle* status, indicating that it remains stationary (hovering). At $t = 1$ UAV-0 begins moving at a constant speed toward the PoI $K = 12$, and continues moving until it successfully reaches and covers the designated PoI.

This paper considers post-disaster emergency scenarios with deterministic communication requirements. Following the settings in [31], [32], a UAV speed range of [20 km/h, 40 km/h] is adopted, along with a mobility model that includes flying and hovering modes ($\bar{v} = 0$).

C. Problem Formulation

The goal of this work is to find a multi-UAV cooperative DCC strategy that specifies how each UAV moves at each time step. Given a strategy, the number of successfully completed DCC tasks within the total execution time T under the given strategy is denoted by

$$\begin{aligned} \Psi &= \sum_{t=0}^T \sum_{k=0}^K b_k \\ &= \sum_{t=0}^T \sum_{k=0}^K \frac{1}{\eta_k^{duration}} \int_{t=\eta_k^{start}}^{t=\eta_k^{start} + \eta_k^{duration}} b_k(t) dt, \end{aligned} \quad (3)$$

$$b_k(t) = \begin{cases} 1, & \text{successful,} \\ 0, & \text{failed,} \end{cases} \quad (4)$$

where $b_k \in \{0, 1\}$, $b_k = 1$ means the PoI k is successfully covered, $b_k = 0$ means failed.

In order to maximize the number of successfully covered tasks Ψ , we need to jointly consider the task start time η_k^{start} , duration $\eta_k^{duration}$ and location $[x^{task}, y^{task}]$ of the PoI, the location $p_i^{uav}(t) = [x_i^{uav}(t), y_i^{uav}(t)]$ of the UAV at each time step, and the spatial geometric relationship between the task location and the CMs. Considering that the process of DCC requires real-time decisions based on the spatiotemporal information of the environment, this paper formulates the DCC optimization problem as a ST-MDP. The optimization objectives and spatiotemporal constraints are expressed as follows:

$$\max \Psi(\eta^{start}, \eta^{duration}, p^{uav}, t, d, \theta), \quad (5)$$

$$\begin{aligned} \text{s.t.} \quad & \text{C1} \quad \eta^{start} \in [0, T], \eta^{duration} \in \{10, 20, 30, 40\}, \\ & \text{C2} \quad x_i^{uav}(t) \in [0, x_{\max}] \wedge y_i^{uav}(t) \in [0, y_{\max}], \\ & \text{C3} \quad d_i(t) \in [0, 1], t \in [0, T], \\ & \text{C4} \quad \theta_i(t) \in (0, 2\pi]. \end{aligned} \quad (6)$$

C1 is the constraint of task start time and duration, which specifies the validity period of the covered tasks. C2 is the

position constraint of the UAV, which requires that the UAV must move within the target area. C3 and C4 specify the flight distance and direction of the UAV in each epoch. The ST-MDP problem suffers from exponential growth in the state-action space as the number of states and actions increases, especially in large-scale scenarios involving spatiotemporal information. It is difficult to list the value of each state, which makes it difficult for model training to converge. To effectively evaluate the value of each state, this paper employs a Transformer to model the nonlinear mapping between states and actions in the environment, estimate the corresponding Q-values, and obtain an approximate optimal policy by maximizing the Q-value.

IV. MULTI-UAV COOPERATIVE DETERMINISTIC COMMUNICATION COVERAGE STRATEGY

In this section, the STA-DRL algorithm is developed to address the DCC optimization problem. This section first introduces the basics of DRL, including the defined state, action, and reward functions. Then, a DRL model trained centrally is employed as a controller to plan the paths for all UAVs.

A. State, Action and Reward

At the epoch t , we define the environment state space, action space, and reward as follows. In the DRL framework, the decision-making process will be executed by the central controller.

1) *State*: As a centralized strategy, the central controller has access to all information regarding the UAV and task status, including the task location $[x^{task}, y^{task}]$, the task's effective coverage duration $[\eta^{start}, \eta^{start} + \eta^{duration}]$, and the UAV's real-time location $p^{uav}(t)$. Therefore, at epoch t , the environment state consists of the task state $s_k^{task}(t)$ and the UAV state $s_i^{uav}(t)$.

$$s(t) = [s_k^{task}(t), s_i^{uav}(t), \dots, s_K^{task}(t), s_N^{uav}(t)], \quad (7)$$

$$s_k^{task}(t) = [x_k^{task}, y_k^{task}, \eta_k^{start}, \eta_k^{duration}, b_k(t), c_k(t)], \quad (8)$$

$$s_i^{uav}(t) = [x_i^{uav}(t), y_i^{uav}(t), o_i(t), u_i(t)], \quad (9)$$

where $c_k(t)$ means the k th PoI is covered by the UAV with index $c_k(t)$ at the k th epoch. $o_i(t)$ means whether UAV_{*i*} is executing the task coverage at epoch t , and $u_i(t)$ indicates that UAV_{*i*} is executing the task with index $u_i(t)$.

2) *Action*: Each UAV needs to select an action from the action space based on the strategy and the current state. The action $a_i(t)$ of the UAV_{*i*} is divided into two parts: the direction $\theta_i(t)$ of the UAV and the flight distance $d_i(t)$ at each epoch. Therefore, the actions of all UAVs are

$$a(t) = [a_0(t), \dots, a_i(t), \dots, a_N(t)]. \quad (10)$$

3) *Reward function*: Upon taking an action derived from the current environment state, the agent immediately receives a reward, and the environment transitions to a new state. To enable the agent to learn an optimal strategy, it is essential to design an appropriate reward function. In general, the reward function is closely related to the optimization goal. Based on (5) and (6), the objective function and constraints are converted into

rewards. The design of the reward function should adhere to the following principles.

- The objective of reinforcement learning is to maximize the cumulative reward value obtained, so the designed reward function should be proportional to the number of tasks Ψ completed.
- The UAV's position should remain within the target area; leaving the area results in a negative reward $r_{out}(t)$.
- Since the number of UAVs is limited, if a task is covered by multiple UAVs simultaneously, a negative reward value $r_{ck}(t)$ is assigned to encourage UAVs to cover different tasks.
- If the drone chooses to cover a task that cannot be completed, it should receive a negative reward value $r_{bk}^-(t)$ to increase the success rate of task coverage. Conversely, if it successfully completes the coverage of a task, it will receive a positive reward value $r_{bk}^+(t)$.

Accordingly, the immediate environmental reward is defined as

$$r(s(t), a(t)) = r_{bk}^-(t) + r_{bk}^+(t) + r_{out}(t) + r_{ck}(t). \quad (11)$$

B. Transformer-Based Q-Value Function Approximator

Typically, the standard Q-learning method stores the obtained state-action pairs and their corresponding Q values in the Q table. The agent selects the action with the maximum Q value from the table based on the current environment state to learn and execute the strategy. However, the large state space resulting from UAV positions and task start times leads to an excessively large Q-table, making the computation of the maximum Q-value highly expensive. Furthermore, many samples may remain unvisited and not updated, preventing the agent from learning effective strategies. To solve the above problems, some works [33] use function approximation methods with stronger generalization capabilities. For example, neural networks, such as deep neural networks (DNNs), are used as Q functions to evaluate the Q value of each state.

Although DNNs can approximate Q-values through iterative training, their network structures have limited capacity to extract key information from the environment, especially complex spatiotemporal features. Inspired by the attention mechanism, the transformer is employed as a Q-value function approximator to enhance the stability of the strategy learned by DQN. The transformer aims to extract critical spatiotemporal features from the environmental state. The proposed STA-DRL architecture is illustrated in Fig. 4. The calculation process of the spatiotemporal feature $\mathbf{Z}_{ST}$ in the environment state is as follows:

$$\mathbf{Z}_{ST} = \mathbf{h}_{all} \cdot \mathbf{W}_{out}, \quad (12)$$

$$\mathbf{h}_{all} = C_{concat}^{(h)}(\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n), \quad (13)$$

$$\mathbf{h}_i = \text{softmax} \left(\frac{\mathbf{Q}_i \cdot \mathbf{K}_j^T}{\sqrt{d_k}} \right) \cdot \mathbf{V}_i, \quad (14)$$

$$\mathbf{Q}_i = s(t) \cdot \mathbf{W}_i^Q,$$

$$\mathbf{K}_i = s(t) \cdot \mathbf{W}_i^K,$$

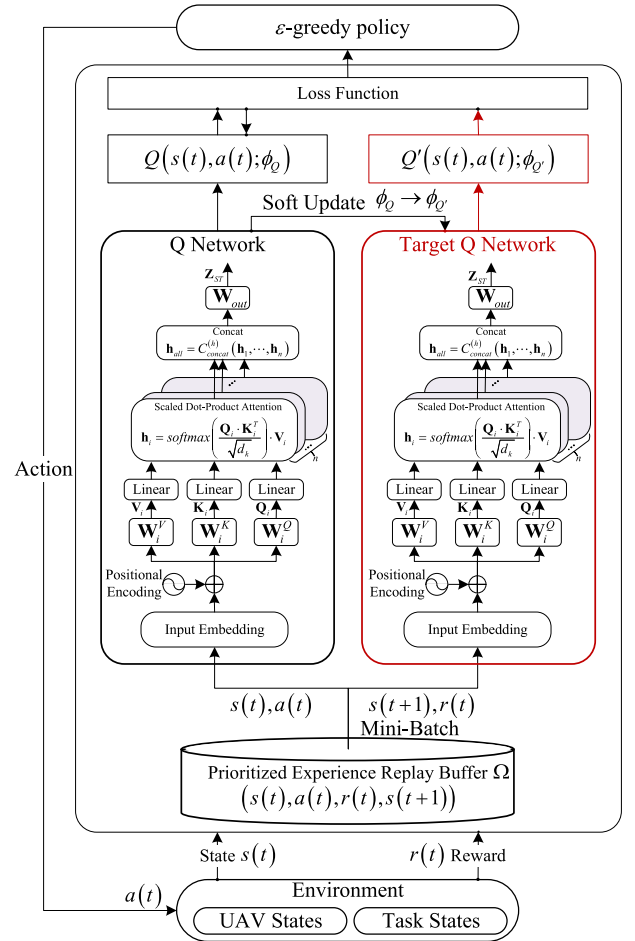


Fig. 3. The framework of the STA-DRL model.

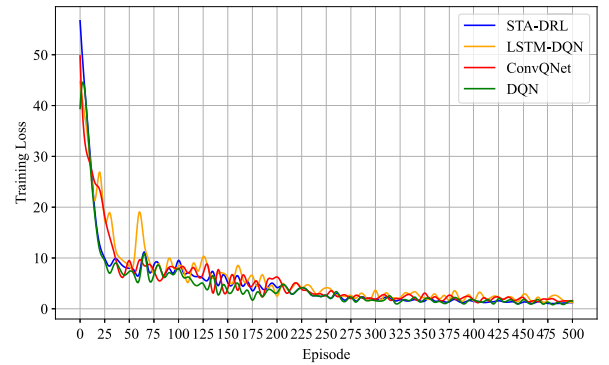


Fig. 4. Training loss curves of different algorithms.

$$\mathbf{V}_i = s(t) \cdot \mathbf{W}_i^V, \quad (15)$$

where $\mathbf{Q}_i$, $\mathbf{K}_i$ and $\mathbf{V}_i$ represent the query, key, and value of the attention matrix, respectively; $\mathbf{h}_i$ means the attention matrix; $s(t)$ means the current input sample; n represents the number of attention heads; $\mathbf{W}_{out}$ represents the learnable weight parameter; $\text{Softmax}(\cdot)$ denotes the Softmax function; d_k represents the dimensions of the matrix.

C. Spatiotemporal Aware Deep Reinforcement Learning

The state-action value function is defined as $Q(s(t), a(t)) = \mathbb{E}(\sum_{t=0}^T \gamma^t r(t) | s(t), a(t))$, where $\gamma \in [0, 1]$ is the discount factor applied to the reward received by the agent at epoch t , and $Q(\cdot)$ is used to evaluate the expected utility of an action given a specific state under the current policy. In order to extract high-dimensional spatiotemporal information in the environment state, we use two transformers with the same structure to serve as the Q network and the target Q network respectively, but with different parameters ϕ_Q and $\phi_{Q'}$.

To improve the sample efficiency and learning stability of reinforcement learning, we adopt a Prioritized Experience Replay Buffer (PERB), denoted as Ω . Unlike uniform sampling, where each experience (sample) has equal probability of being selected, PERB assigns higher sampling priority to experiences with larger Temporal Difference errors (TD-errors), as these samples are considered to carry more learning value. Formally, the priority p_i of each experience i is defined as

$$p_i = |\delta_i| + \varepsilon, \quad (16)$$

where δ_i is the TD-error of experience i , and ε is a small positive constant that ensures non-zero priority. Based on these priorities, the sampling probability for each experience is computed as

$$P(i) = \frac{p_i^\mu}{\sum_k p_k^\mu}, \quad (17)$$

where $\mu \in [0, 1]$ controls the degree of prioritization. To correct the bias introduced by non-uniform sampling, Importance Sampling Weights (ISWs) are used during training, calculated as

$$w_i = \left(\frac{1}{\text{len}(\Omega) P(i)} \right)^\beta, \quad (18)$$

where $\text{len}(\Omega)$ is the size of Ω and $\beta \in [0, 1]$ controls the amount of bias correction. These ISWs are normalized to prevent instability during optimization.

By focusing on experience with higher TD-errors, the PERB enables the agent to prioritize informative experiences, thereby accelerating convergence and improving overall performance. The parameter update process of STA-DRL is shown in Algorithm 1. From lines 4 to 7, the agent outputs all uav actions in a given state. In line 8, the difference between the current Q value and the target Q value is calculated, which is called the TD-error $\delta(t) = r(t) + \gamma \max_{a'} Q'(s(t+1), a'; \phi_{Q'}) - Q(s(t), a(t); \phi_Q)$, to update the Q network. In line 9, all transition states are stored in Ω . From lines 10 to 18, B mini-batch samples are sampled from Ω to train the transformer. From lines 14 to 15, the loss is computed using the function $L(\phi_Q) = \frac{1}{B} \sum_i \omega_i \cdot (Q'(i; \phi_{Q'}) - Q(s_i, a_i; \phi_Q))^2$. In line 16, the gradient descent method is used to update the parameters ϕ_Q of the Q network, where $\alpha \in [0, 1]$ is the learning rate and ∇_{ϕ_Q} is the gradient function with respect to ϕ_Q . In line 17, the target Q network parameters $\phi_{Q'}$ are updated through a soft update mechanism [34] to gradually approach the Q network parameters ϕ_Q . In line 18, to achieve a balance between exploration and exploitation, ε

TABLE II
SETTING AND HYPER-PARAMETERS USED IN OUR SIMULATION

Parameter	Default value
Target region S	$30 \times 30(\text{km})$
Number of PoIs K	50,75,100
Task start time η_k^{start}	$(0, 180]$ (epoch)
Task duration η_k^{duration}	$\{10, 20, 30, 40\}$ (epoch)
Number of CHs M	4
Number of CMs N	4,8,12,16,20,24
CM's flight speed $\bar{v}$	$[20 \text{ km/h}, 40 \text{ km/h}]$
CM's initial position	$(15 \text{ km}, 15 \text{ km})$
CM's coverage range R	1,2,3 (km)
Total epochs T	180
Episodes T_e	500
Learning rate α	$\{0.001, 0.01\}$
Number of mini-batch B	32
Exploration rate $\varepsilon_{\min}$	0.01
Epsilon decay rate $\varepsilon_{\text{decay}}$	$\{0.98, 0.995\}$
Discount factor γ	$\{0.9, 0.99\}$

is dynamically adjusted. Finally, after T_e episodes of iterative training, the optimal strategy π^* is obtained.

V. SIMULATIONS AND RESULTS

In this section, we build a UAV deterministic network to verify the effectiveness of the proposed method. Next, we introduce the simulation setup used throughout our experiments. Then, we investigate the training convergence behavior of each algorithm and analyze the impact of hyperparameters on model performance. Finally, we compare the proposed method with both baseline and state-of-the-art approaches. These evaluations collectively demonstrate the robustness and advantages of our method under various conditions.

A. Simulation Setup

The simulation hyper-parameter settings are primarily informed by [22], [35], [36], the limited energy capacity of UAVs makes them better suited to act as communication relays for short-duration coverage tasks in post-disaster rescue and recovery scenarios. In our simulation, we set the target region to be a square area with a size of 30×30 units, where each unit corresponds to 1 km. We randomly deploy K PoIs in this region, where the task start time and continuous coverage time of each PoI are $\eta_k^{\text{start}} \in (0, 180]$ and $\eta_k^{\text{duration}} \in \{10, 20, 30, 40\}$, respectively. The CM's maximum flight time is set to 3 hours, and the total task execution time is 180 epochs, 1 epoch = 1 minute. The CM's flight speed $\bar{v}$ is set within the range of $[20 \text{ km/h}, 40 \text{ km/h}]$. Our simulation runs were performed with Pytorch 2.0 and Python 3.9 on a Windows 10 with 1 NVIDIA GeForce RTX 4070 SUPER GPUs. We trained the proposed STA-DRL models for 500 episodes, each of which has 180 epochs. After training, we tested it for a period of $T = 180$ epochs. Table II summarizes various key experimental parameters.

B. Training Convergence

In this simulation, we set the number of CMs $N = 16$, the number of PoIs $K = 50$, the CM's flighted speed $\bar{v} = 20 \text{ km/h}$ and coverage range $R = 1 \text{ km}$.

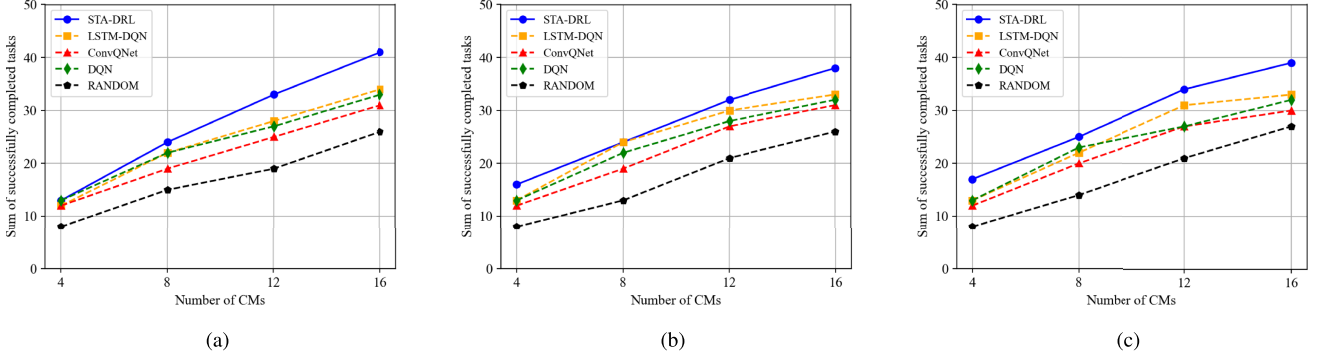


Fig. 5. The impact of the number of CMs on the number of completed coverage tasks, with $\bar{v} = 20$ km/h and $K = 50$. (a) CM cover range $R = 1$ (km). (b) CM cover range $R = 2$ (km). (c) CM cover range $R = 3$ (km).

Algorithm 1: STA-DRL.

```

1: initialization:  $\varepsilon = 1$ ,  $Q(s, a; \phi_Q)$  and  $Q'(s, a; \phi_{Q'})$ ;
2: for  $t_e = 0$  to  $T_e$  (the number of episodes) do
3:   Let  $t = 0$ , and get the initial state  $s(t)$ ;
4:   for  $t = 0$  to  $T$  do
5:     Take action  $a(t)$  with  $\varepsilon$ -greedy policy at  $s(t)$ ;
6:     Obtain the reward  $r(t)$ ;
7:      $s(t)$  transitions to  $s(t + 1)$ ;
8:     Compute TD-error  $\delta(t)$ ;
9:     Store a  $(s(t), a(t), r(t), s(t + 1))$  in the  $\Omega$ ;
10:    Randomly extract  $B$  mini-batch samples from  $\Omega$ ;
11:    for  $b = 1$  to  $B$  do
12:      Obtain  $Q(b)$  and  $Q'(b)$ ;
13:    end for
14:    Compute ISW:  $w_i = (\frac{1}{\text{len}(\Omega)P(i)})^\beta$ ;
15:     $L(\phi_Q) = \frac{1}{B} \sum_i w_i \cdot (Q'(i; \phi_{Q'}) - Q(s_i, a_i; \phi_Q))^2$ ;
16:     $\phi_Q \leftarrow \phi_Q - \alpha \nabla_{\phi_Q} L(\phi_Q)$ ;
17:    soft update target Q-network  $Q'(\cdot)$ ;
18:     $\varepsilon \leftarrow \max(\varepsilon_{\min}, \varepsilon_{\text{decay}} \cdot \varepsilon)$ 
19:  end for
20: end for
21: output: The optimal policy  $\pi^*$ 

```

Fig. 5 illustrates the convergence process of all algorithms. Significant fluctuations are observed at the beginning, primarily due to the inaccuracy of the initial function approximation, which results in high loss values during training. During the mid-stage of training, the training losses of all algorithms steadily decrease over time, indicating that each algorithm is actively exploring and learning improved policies. Before the 275-th episode, the DQN, LSTM-DQN, and ConvQNet algorithms had already converged, which can be attributed to their relatively simple network structures that are easier to train and fit. In contrast, the STA-DRL algorithm begins to converge around episode 350, primarily due to its more complex network structure, which requires a longer training period to achieve effective convergence.

TABLE III

IMPACT OF DIFFERENT HYPERPARAMETER SETTINGS ON PERFORMANCE

α	γ	$\varepsilon_{\text{decay}}$	Ψ	$\Psi(\text{failure})$
0.001	0.9	0.995	41	39
0.001	0.9	0.98	42	39
0.001	0.99	0.995	41	39
0.001	0.99	0.98	38	38
0.01	0.9	0.995	39	37
0.01	0.9	0.98	39	37
0.01	0.99	0.995	37	38
0.01	0.99	0.98	38	37

C. Finding Appropriate Hyperparameters

Our goal is to identify suitable hyperparameters to enhance model performance by varying the learning rate, discount factor, and epsilon decay rate. We consider 16 CMs, 50 PoIs and coverage range $R = 1$ km. From Table III, it was found that setting $\alpha = 0.001$, $\gamma = 0.9$ and $\varepsilon_{\text{decay}} = 0.98$ yields the best model performance, regardless of whether CM failures occur or not. Specifically, increasing α from 0.001 to 0.01 leads to a decrease of approximately 2 to 3 in the number of successfully covered tasks (Ψ) by STA-DRL. This degradation is likely due to the instability introduced by a higher learning rate, which can cause the model to overshoot optimal solutions. Furthermore, the performance of STA-DRL is slightly improved when γ is reduced from 0.99 to 0.9. In contrast, the performance remains largely unaffected by small changes in the epsilon decay rate.

D. Comparing With Baseline Approaches

Since the action space in this work is discrete, DQN-based models are used as baseline methods. In the following, we compare the proposed STA-DRL algorithm with four other algorithms.

- 1) *DQN*: The Q-network in this model is implemented using a DNN, which possesses strong nonlinear function approximation capabilities. This enables the DQN to capture the complex, nonlinear relationships between states and actions [33].
- 2) *ConvQNet*: This model integrates Convolutional Neural Networks (CNN) with DQN, leveraging the ability of CNNs to effectively extract features from high-dimensional environmental state while preserving spatial

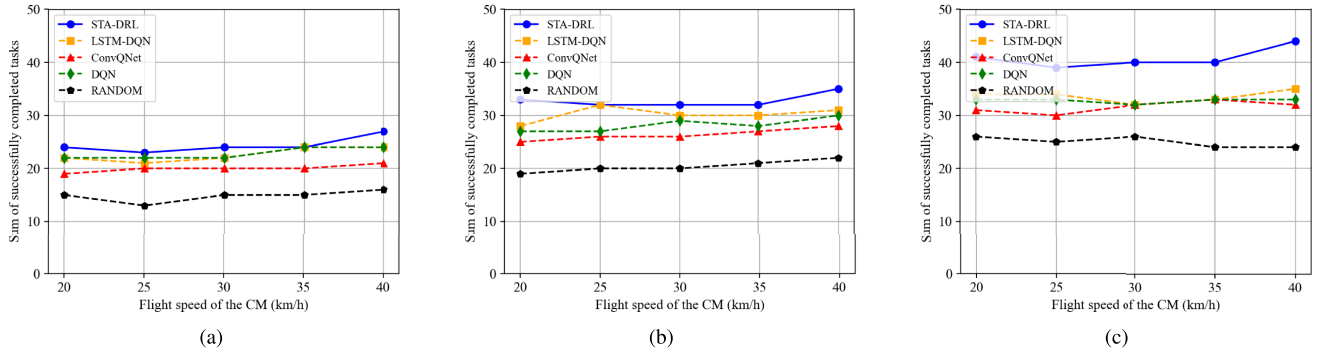


Fig. 6. The impact of the flight speed of CM on the number of successfully completed tasks, with $R = 1$ km. (a) Number of CMs $N = 8$. (b) Number of CMs $N = 12$. (c) Number of CMs $N = 16$.

information. This combination enhances the model's generalization capability [37].

- 3) *LSTM-QNet*: The primary advantage of combining LSTM with DQN is their ability to address temporal dependencies and long-term relationships, particularly in applications that require long-term decision-making and operate in dynamic environments [38].
- 4) *RANDOM*: This is an extension of a basic random strategy, where the actions of all CMs are selected randomly, with the additional constraint that all CMs must remain within a specified target area.

In simulation scenario 1, we show the impact of the CMs coverage range and the number of CMs on the number of successfully completed tasks in Fig. 5. We fixed $\bar{v} = 20$ km/h, while varying the number of CMs from 4 to 16. We can make the following observation from Fig. 5.

1) As the coverage increases, the number of tasks completed by the same number of CMs rises to a certain extent. This indicates that, in the scenario of point-to-point coverage tasks with time constraints, the size of the coverage area has minimal impact on the performance of the method.

2) The STA-DRL consistently outperforms other methods, particularly as the number of CMs increases, where the advantage of STA-DRL becomes more significant. As shown in Fig. 5(a), when the CM's coverage is 1 km, the total number of PoIs in the target area is 50, and the number of CMs is 16, STA-DRL completes 41 tasks within the given time T , while other methods such as LSTM-DQN, ConvQNet, and DQN complete 34, 31, and 33 tasks, respectively. When the number of CMs is 4, except for the RANDOM method, the number of coverage tasks completed by the other methods is similar. In comparison to the temporal features extracted by LSTM and the spatial features extracted by CNN, STA-DRL employs a transformer to extract more effective spatiotemporal features from the environmental state.

3) Compared to other methods, RANDOM performs poorly, which shows that the random strategy cannot effectively solve the coverage problem in this scenario.

In simulation scenario 2, Fig. 6 presents the impact of the flight speed of CM on the number of successfully completed tasks. In Fig. 6(a)–(c), the total number of CMs is set to 8, 12, and 16

respectively. In each sub-scenario, the speed range of CMs is set from 20 to 40 km/h. We can make the following observation from Fig. 6.

1) As $\bar{v}$ increases from 20 km/h to 35 km/h, the performance of all algorithms remains largely unchanged. When $N = 8, 12$, and 16, with $\bar{v}$ is fixed at 40 km/h, the performance gap between STA-DRL and LSTM-DQN is 3, 4, and 9, respectively. When $\bar{v} = 20$ km/h, the performance gap between STA-DRL and LSTM-DQN is 2, 5, and 7, respectively. STA-DRL has always been better than other methods. This shows that the extracted spatiotemporal features can improve the performance of the model.

2) As shown in Fig. 6(a) and Fig. 6(b), except for STA-DRL, other methods are not sensitive to changes in the speed of CMs. However, for STA-DRL, the number of completed tasks will increase only when $\bar{v}$ increases from 35 km/h to 40 km/h. This shows that when the speed reaches a certain level, it will affect the performance of the algorithm.

3) Fig. 6(c) illustrates that, apart from RANDOM, the performance of other methods is similar. LSTM-DQN, ConvQNet, and DQN exhibit signs of weakness, whereas STA-DRL maintains strong performance, demonstrating its ability to effectively extract key spatiotemporal information from the environmental state.

In simulation scenario 3, we show the impact of the number of CMs and PoIs on the number of successfully completed tasks in Fig. 7. Compared to simulation scenario 1, we fix $R = 1$ km, $\bar{v} = 20$ km/h, while varying N from 8 to 24. In Scenarios 1.a and 1.b, the total number of PoIs is set to $K = 75$ and $K = 100$, respectively, while the simulation time T remains unchanged. We can make the following observation from Fig. 7.

1) As the number of CMs increases, the number of coverage tasks successfully completed by the UAV cluster also increases. As shown in Fig. 7(a), when $N = 24$, STA-DRL completes 54 tasks, while LSTM-DQN, ConvQNet, DQN, and RANDOM complete 49, 41, 42, and 32 tasks, respectively. This demonstrates the effectiveness of the STA-DRL method. Notably, the performance of LSTM-DQN is slightly below that of STA-DRL, suggesting that temporal information in the environment state plays a more crucial role than spatial information.

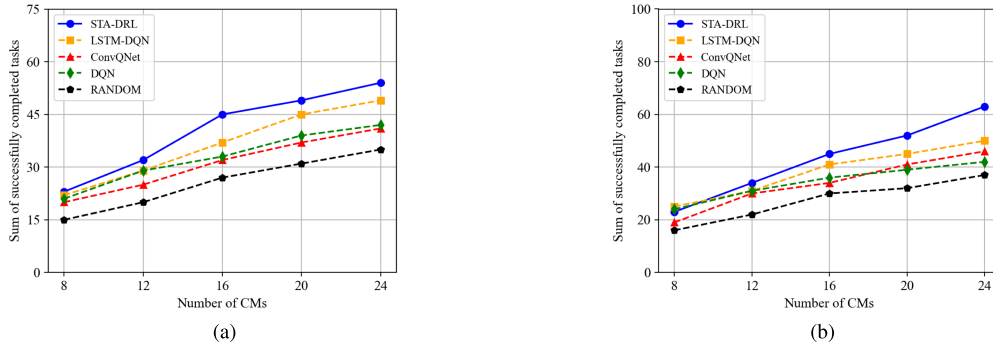


Fig. 7. The impact of the number of CMs and PoIs on the number of completed coverage tasks, with $\bar{v} = 20$ km/h and $R = 1$ km. (a) Number of PoIs $K = 75$. (b) Number of PoIs $K = 100$.

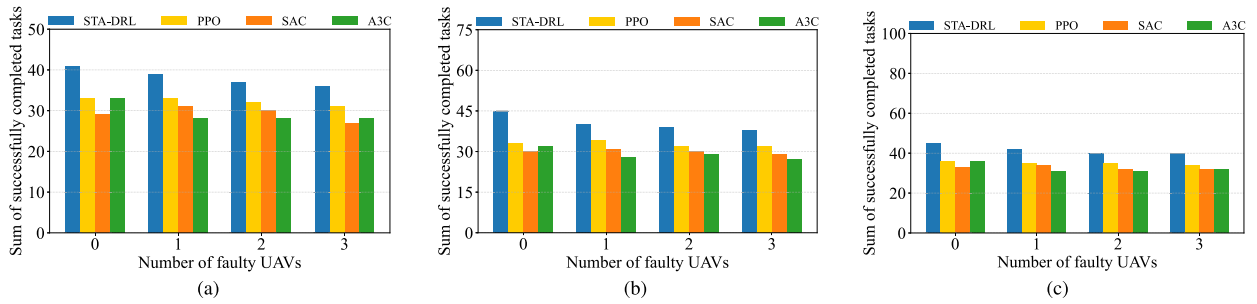


Fig. 8. The impact of the number of faulty UAVs on the number of successfully completed tasks, with $N = 16$, $\bar{v} = 20$ km/h and $R = 1$ km. (a) Number of PoIs $K = 50$. (b) Number of PoIs $K = 75$. (c) Number of PoIs $K = 100$.

2) By comparing the results in Fig. 7(a) and Fig. 7(b), we observe that as the total number of PoIs increases from 75 to 100, the number of coverage tasks completed by the drone cluster increases monotonically. For example, when $N = 24$, the total number of tasks covered by STA-DRL increases from 54 to 63, while the number of tasks covered by ConvQNet increases from 41 to 46. This implies that the increase in the number of PoIs provides more task options for the CMs. The centralized controller, with the objective of maximizing the number of completed tasks, can select more optimal paths for each CM.

3) Another interesting observation is that as the number of CMs increases, the performance gap between the proposed STA-DRL method and other methods widens, further validating that the method is well-suited for larger UAV clusters.

E. Comparing With State-of-The-Art Approaches

Considering that CMs may fail to operate during emergency rescue missions, we designed a simulation scenario with random failures to evaluate the robustness and practical applicability of the proposed method under such conditions. Specifically, during the simulation, 0 to 3 drones were randomly selected to fail at random time points. We use three approaches to compare with our algorithm.

- 1) PPO [27]: It is a policy gradient-based method that uses a clipped surrogate objective to limit the update between the

new and old policies, aiming to balance training stability and sample efficiency.

- 2) A3C [24]: A3C accelerates the training process by leveraging multiple asynchronous worker threads running in parallel. It combines the strengths of the actor-critic architecture and introduces an advantage function to reduce estimation variance and enhance learning stability.
- 3) SAC [8]: SAC is a maximum entropy-based reinforcement learning approach, which optimizes a combined objective of reward and entropy to balance exploration and exploitation, thereby enhancing training stability and sample efficiency.

During testing, all approaches run 180 epochs in an episode. We conduct three sets of simulations by varying K and number of faulty CMs, with $N = 16$, $\bar{v} = 20$ km/h, and $R = 1$ km. We present their results by the number of tasks successfully completed.

From Fig. 8(a)–(c), it can be observed that the performance of all approaches declines as the number of faulty CMs increases in each episode. This degradation is primarily due to the inability of failed CMs to participate in subsequent task coverage. Among all the evaluated methods, the proposed approach exhibits superior robustness, with PPO consistently achieving the second-best performance.

It is observed that, when $N = 16$, increasing K does not lead to a significant improvement in the number of completed tasks. This observation indicates that the multi-UAV cooperative DCC

strategies generated by all approaches are already near-optimal under this configuration.

A3C exhibits inferior performance compared to other approaches, primarily because it is an on-policy algorithm; each policy update relies exclusively on trajectories generated by the current policy, which limits sample efficiency. In the scenario considered in this work, where the state space is high-dimensional and historical experience is critical, this limitation significantly impacts performance. Therefore, A3C is less suitable for addressing the DCC problem presented in this paper.

VI. CONCLUSION

In this article, this paper proposes an STA-DRL approach for multi-UAV cooperative control in dynamic environments. The proposed method formulates the multi-task DCC problem as an ST-MDP and introduces a novel reward function that jointly captures the spatiotemporal relationships between UAV and task states, while considering task validity constraints. Decision-making is further enhanced through the integration of dual Transformer structures. Extensive simulation results show that STA-DRL consistently outperforms four baseline methods: DQN, ConvQNet, LSTM-QNet, and RANDOM. This superiority is reflected in the number of tasks successfully completed. Furthermore, UAV failure scenarios are considered in the simulations, and the proposed method is compared with three state-of-the-art approaches. The results demonstrate that it exhibits strong robustness under such conditions.

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